Groups and Coding

群与编码

Groups and Coding

- Coding of Binary Information and Error Detection

（二元信息编码与差错检测）

- Decoding and Error Correction

（译码与纠错）

Coding of Binary Information and Error Detection

Encoding function（编码函数）

Hamming distance（汉明距离）

Group Codes （群码）

Parity check matrix (奇偶校验矩阵)

Coding theory

In today's modern world of communication, data items are constantly being transmitted from point to point. The basic problem in transmission of data is that of receiving the data as sent and not receiving a distorted piece of data.

Coding theory has developed techniques for introducing redundant information(引入冗余信息) in transmitted data that help in detecting, and sometimes in correcting, errors.

Unit of information
信息单位

Message(消息) is a finite sequence of characters from a finite alphabet B = {0, l }

Word(字) is a sequence of m 0's and l 's.

Groups

The set B is a group under the binary operation + (mod 2 addition)

It follows that $B^m = B \times B \times \ldots \times B$ (m factors) is a group under the operator $\oplus$ defined by

$$(x_1, x_2, \ldots, x_m) \oplus (y_1, y_2, \ldots, y_m)$$
$$= (x_1 + y_1, x_2 + y_2, \ldots, x_m + y_m)$$

An element in B^m will be written as $(b_1, b_2, \ldots, b_m)$ or more simply as $b_1 b_2 \ldots b_m$

元素$(b_1,b_2,\ldots,b_m)$，简写为 $b_1b_2\ldots b_m$

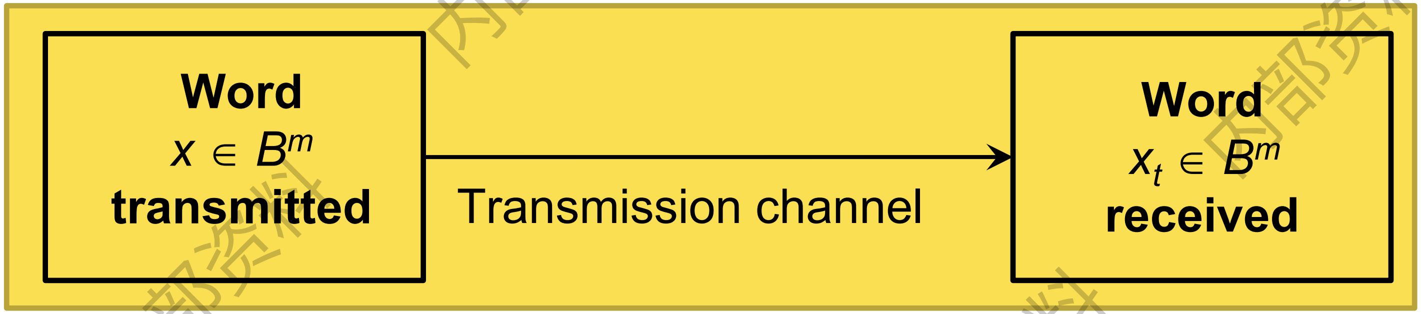
例:$(1,0,0,1,0,0)\oplus(1,1,0,0,1,1) = (1+1,0+1,0+0,1+0,0+1,0+1)$

$= （0，1，0，1，1，1）$

$= 0\ 1\ 0\ 1\ 1\ 1$

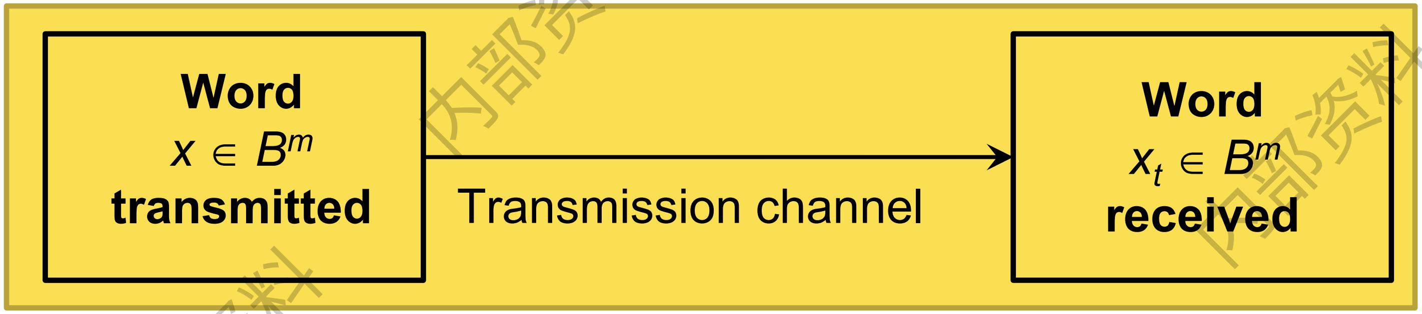
Transmission channel and Noise 传输信道和噪声

An element $x \in B^m$ is sent through the transmission channel and is received as an element $x_t \in B^m$.



Transmission channel and Noise
传输信道和噪声

Noise(噪声) in the transmission channel may cause a 0 to be received as a l, or vice versa, lead $x \neq x_t$



对 *Noise* 对 寸（✕）

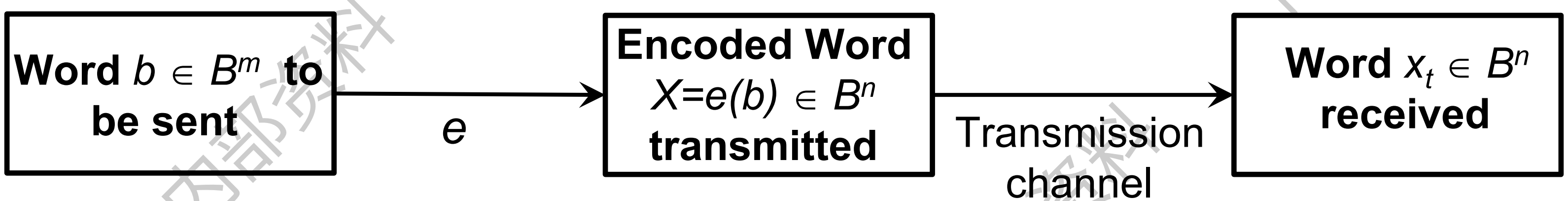
11001011010111110111001 → 11001011010111110111001

如何检测接收的消息有误码？

Encoding function - 编码函数

Choose: an integer $n>m$ and a one-to-one function e: $B^m \rightarrow B^n$

- e is called an (m, n) *encoding function*, representing every word in B^m as a word in B^n.
- If $b \in B^m$, then $e(b)$ is called the *code word*(码字) representing b.



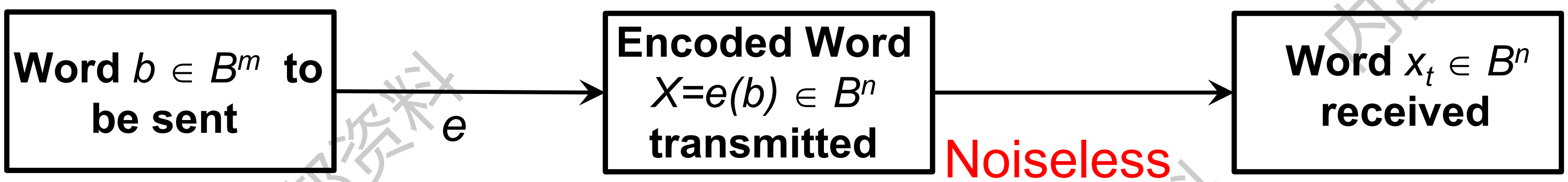
$b=b_1b_2...b_m$

$e（b）=b_1b_2...b_m\ x_1\ x_2...\ x_{n-m}$

Encoding function

If the transmission channel is noiseless, then $x_t = x$ for all x in B^n.

In this case $x = e(b)$ is received for each $b \in B^m$, and since e is a known function, b may be identified.

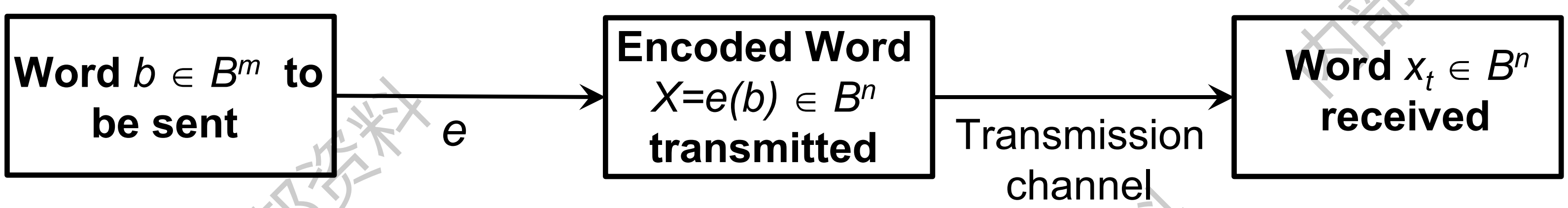


$b=b_1b_2\ldots b_m$ $e(b)=b_1b_2\ldots b_m\, x_1\, x_2\ldots\, x_{n-m}$ $x_t=b_1b_2\ldots b_m\, x_1\, x_2\ldots\, x_{n-m}$

Encoding function

In general, errors in transmission do occur.

We will say that the code word（代码字）$x = e(b)$ has been transmitted with *k or fewer errors* if x and x_t differ in at least 1 but no more than k positions.



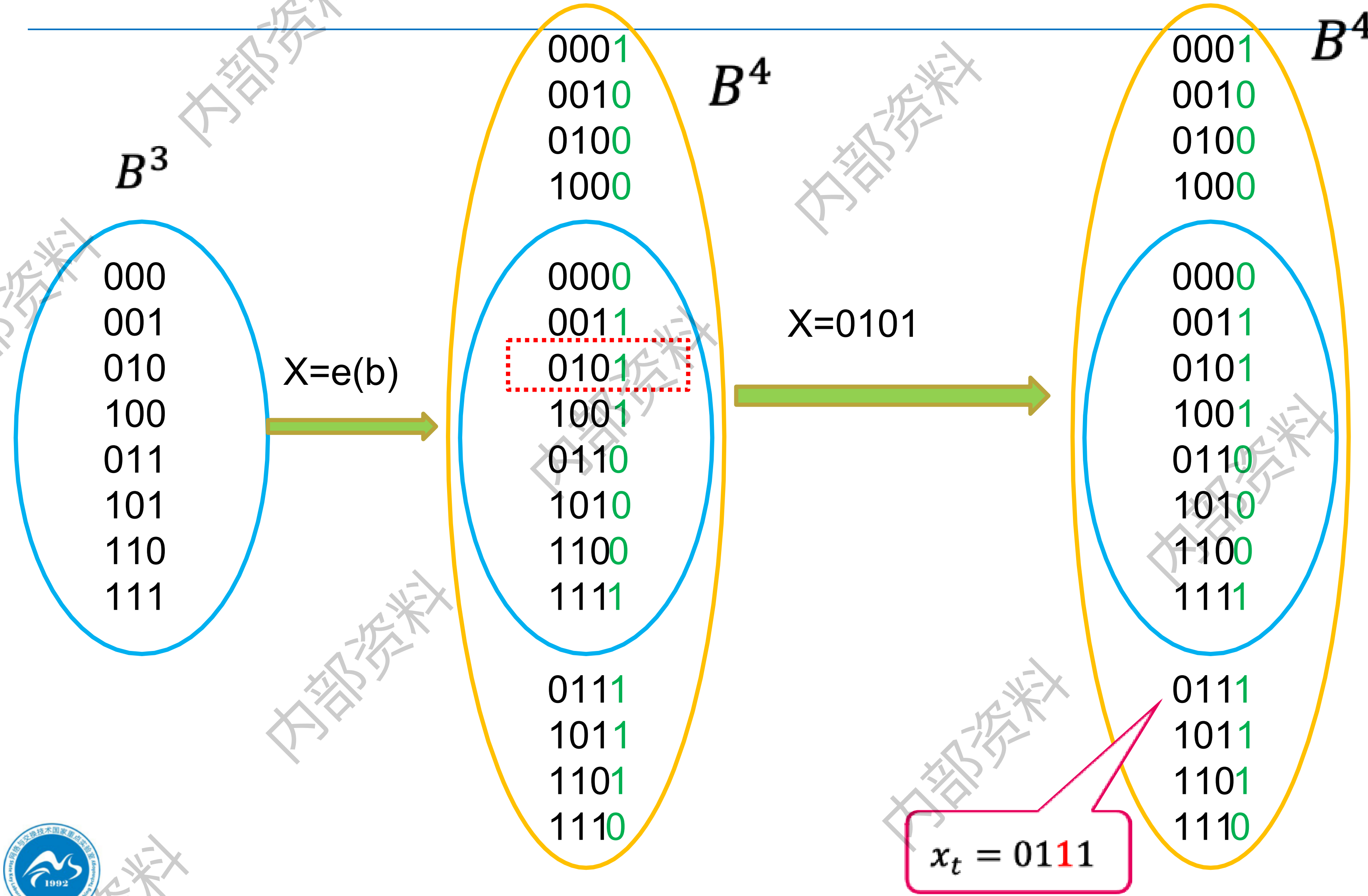
1100101101011111011110001001 11001011000111110011011001

1至k位错误

Error detect – 差错检测

Let $e: B^m \to B^n$ be an (m, n) encoding function.

e *detects k or fewer errors* if whenever $x = e(b)$ is transmitted with k or fewer errors, then x_t is not a code word (thus x_t could not be x and therefore could not have been correctly transmitted).


B^3
000
001
010
100
011
101
110
111
X=e(b)
B^4
0001
0010
0100
1000
0000
0011
0101
1001
0110
1010
1100
1111
0111
1011
1101
1110
X=0101
B^4
0001
0010
0100
1000
0000
0011
0101
1001
0110
1010
1100
1111
0111
1011
1101
1110
$x_t = 0111$

Error detect – 差错检测

For $x \in B^n$, the number of 1's in x is called the weight(权) of x and is denoted by $|x|$

Example 1:

Find the weight of each of the following words in B^5.

x=01000	$\|x\|$=1
x=11100	$\|x\|$=3
x=00000	$\|x\|$=0
x=11111	$\|x\|$=5

Example 2
parity check code –奇偶校验码

The following encoding function $e: B^m \to B^{m+1}$ is called the *parity (m, m+1) check code*: If $b = b_1 b_2 \ldots b_m \in B^m$, define

$$e(b) = b_1 b_2 \ldots b_m\, b_{m+1}$$

Where

$$b_{m+1} = \begin{cases} 0 & \text{if } |b| \text{ is even} \\ 1 & \text{if } |b| \text{ is odd} \end{cases}$$

Example 2
parity check code

Let $m = 3$. Then

$e(000) = 0000$

$e(001) = 0011$

$e(010) = 0101$

$e(011) = 0110$

$e(100) = 1001$

$e(101) = 1010$

$e(110) = 1100$

$e(111) = 1111$

Suppose now that $b = 111$.

$x = e(b) = 1111$ $\quad$ $x_t = 1011$

Example 3

Consider the $(m, 3m)$ encoding function e: $B^m \to B^{3m}$.

If $b = b_1 b_2 \ldots b_m \in B^m$, define

$$e(b) = b_1 b_2 \ldots b_m b_1 b_2 \ldots b_m b_1 b_2 \ldots b_m$$

Example 3

$$
\left.\begin{aligned}
e(000) &= 000000000\\
e(001) &= 001001001\\
e(010) &= 010010010\\
e(011) &= 011011011\\
e(100) &= 100100100\\
e(101) &= 101101101\\
e(110) &= 110110110\\
e(111) &= 111111111
\end{aligned}\right\}\ \text{code word}
$$

Suppose now that b = 011

Then e(011) = 011011011.

111111011

Assume now we receive the word 011111011. This is not a code word, so we have detected the error.

011101011

例：4个校验位

1	0	1	1	0	0	0	1	1	0	1	0	0	0
1	0	1	1	0	1	0	1	1	0	1	0	0	0
b1	b2	b3	b4	b5	b6	b7	b8	b9	b10	b11	b12	b13	b14
b1	b2	b3	b4	b5	b6	b7	b8	b9	b10	b11	b12	b13	b14
b1	b2	b3	b4	b5	b6	b7	b8	b9	b10	b11	b12	b13	b14
b1	b2	b3	b4	b5	b6	b7	b8	b9	b10	b11	b12	b13	b14

$$b11 = b1 \oplus b3 \oplus b5 \oplus b7 \oplus b9$$
$$b12 = b2 \oplus b3 \oplus b6 \oplus b7 \oplus b10$$
$$b13 = b4 \oplus b6 \oplus b7$$
$$b14 = b8 \oplus b9 \oplus b10$$

如何衡量不同消息间的差异程度？

Hamming distance – 汉明距离

Let x and y be words in B^m. The *Hamming distance* $\delta(x, y)$ between x and y is the weight, $|x \oplus y|$, of $x \oplus y$.

The distance between $x = x_1x_2\ldots x_m$ and $y = y_1y_2\ldots y_m$ is the number of various of i such that $x_i \neq y_i$, that is, the number of positions in which x and y differ.

Using the weight of $x \oplus y$ is a convenient way to count the number of different positions.

$$
\begin{aligned}
x &= 010011 \\
y &= 110110 \\
x \oplus y &= 100101
\end{aligned}
$$

Example 4

Find the distance between x and y:

(a) $x = 110110,\ y = 000101$

(b) $x = 001100,\ y = 010110.$

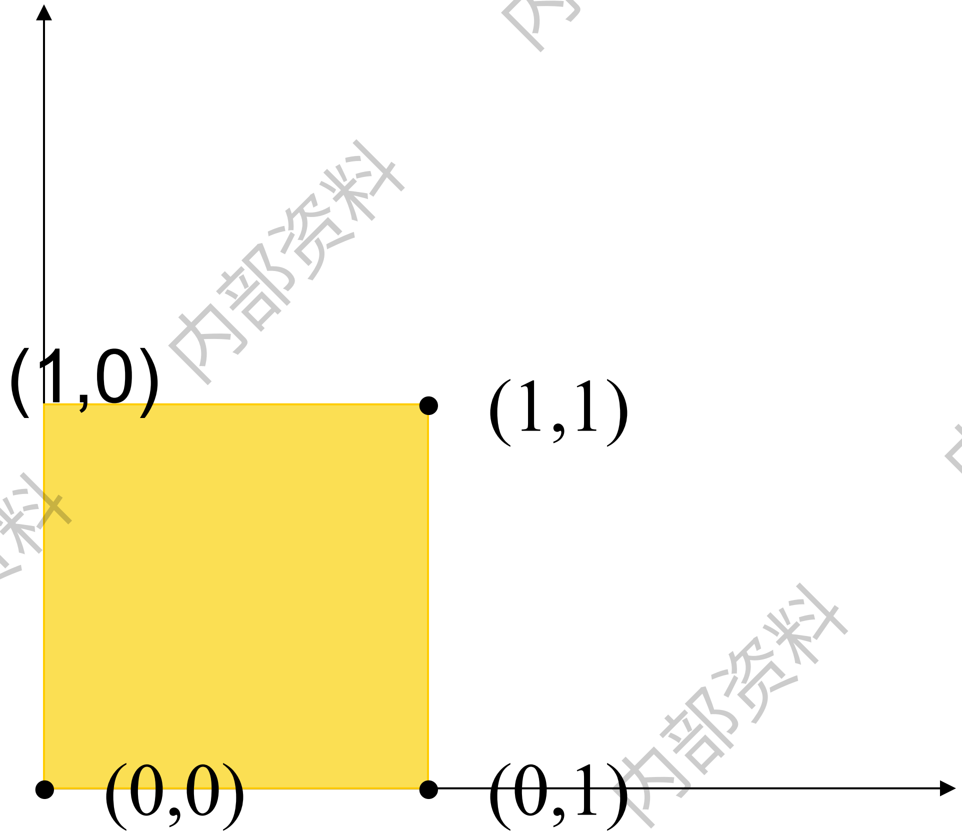
Solution

(a) $x \oplus y = 110011$, so $|x \oplus y| = 4$

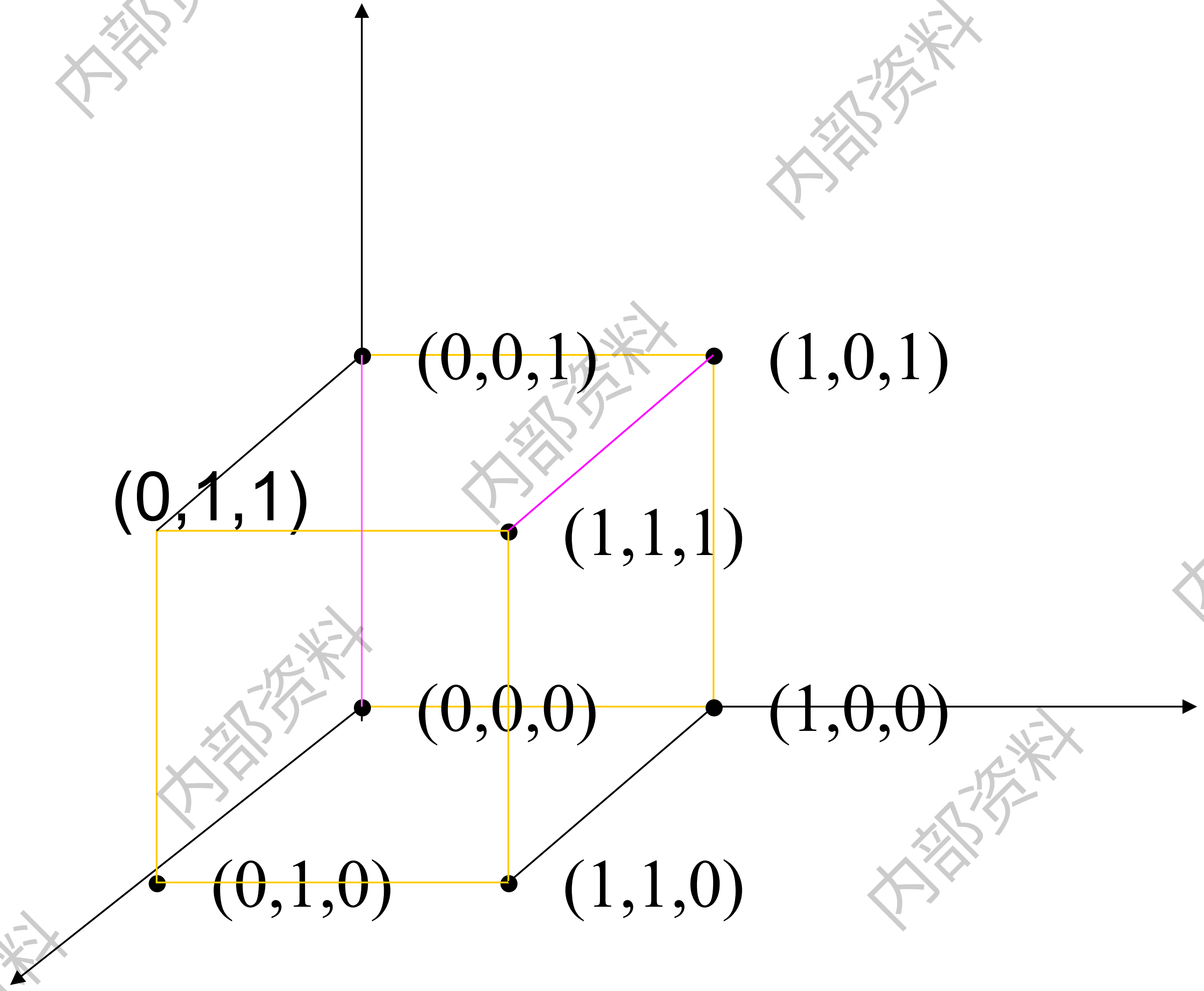
$$x = 001100$$
$$y = 010110$$

(b) $x \oplus y = 011010$, so $|x \oplus y| = 3$

(00,10,01,11)
(00,11)



$B^m(0,1)\text{->}B^n(000,111)$



Theorem 1

properties of distance function

距离函数的性质

Let x, y, and z be elements of B^m. Then

(a) $\delta(x, y) = \delta(y, x)$

(b) $\delta(x, y) \geq 0$

(c) $\delta(x, y) = 0$ if and only if $x = y$

(d) $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$

Proof of (d)

$$|x \oplus y| \leq |x| + |y|; \qquad a \oplus a = \mathbf{0}$$

$$\delta(x, y) = |x \oplus y| = |x \oplus \mathbf{0} \oplus y|$$

$$= |x \oplus z \oplus z \oplus y|$$

$$\leq |x \oplus z| + |z \oplus y|$$

Minimum distance
最短距离

The *minimum distance* of an encoding function $e: B^m \to B^n$ is the minimum of the distances between all distinct pairs of code words; that is,

$$\min\{\delta(e(x), e(y)) \mid x, y \in B^m\}$$

Example 5

Consider the following (2, 5) encoding function *e*:

$$e(00) = 00000$$
$$e(01) = 01110$$
$$e(10) = 10011$$
$$e(11) = 11111$$

$\delta_{min} = 2$

Theorem 2
-检测K位错误

An (m, n) encoding function $e: B^m \to B^n$ can detect k or fewer errors

if and only if

its minimum distance is at least $k + 1$.

$\delta_{\min} \geq k+1$

Proof

$\Leftarrow$ the minimum distance is at least $k+1$

Let $b \in B^m$, and let $x = e(b) \in B^n$ be the code word representing b.

x is transmitted and is received as x_t. If x_t were a code word different from x, then $\delta(x, x_t) \geq k+1$, so x would be transmitted with $k+1$ or more errors.

Thus, if x is transmitted with k or fewer errors, then x_t cannot be a code word.

This means that e can detect k or fewer errors.

证充分性：思路--- $e(b)$最小距离大于等于k+1，如果$\delta(x，x_t) \leq k$时，说明x_t一定不是正确的代码字。

Proof

e can detect k or fewer errors $\Rightarrow$

Suppose that the minimum distance between code words is $r \le k$

Let x and y be code words with $\delta(x, y) = r$.

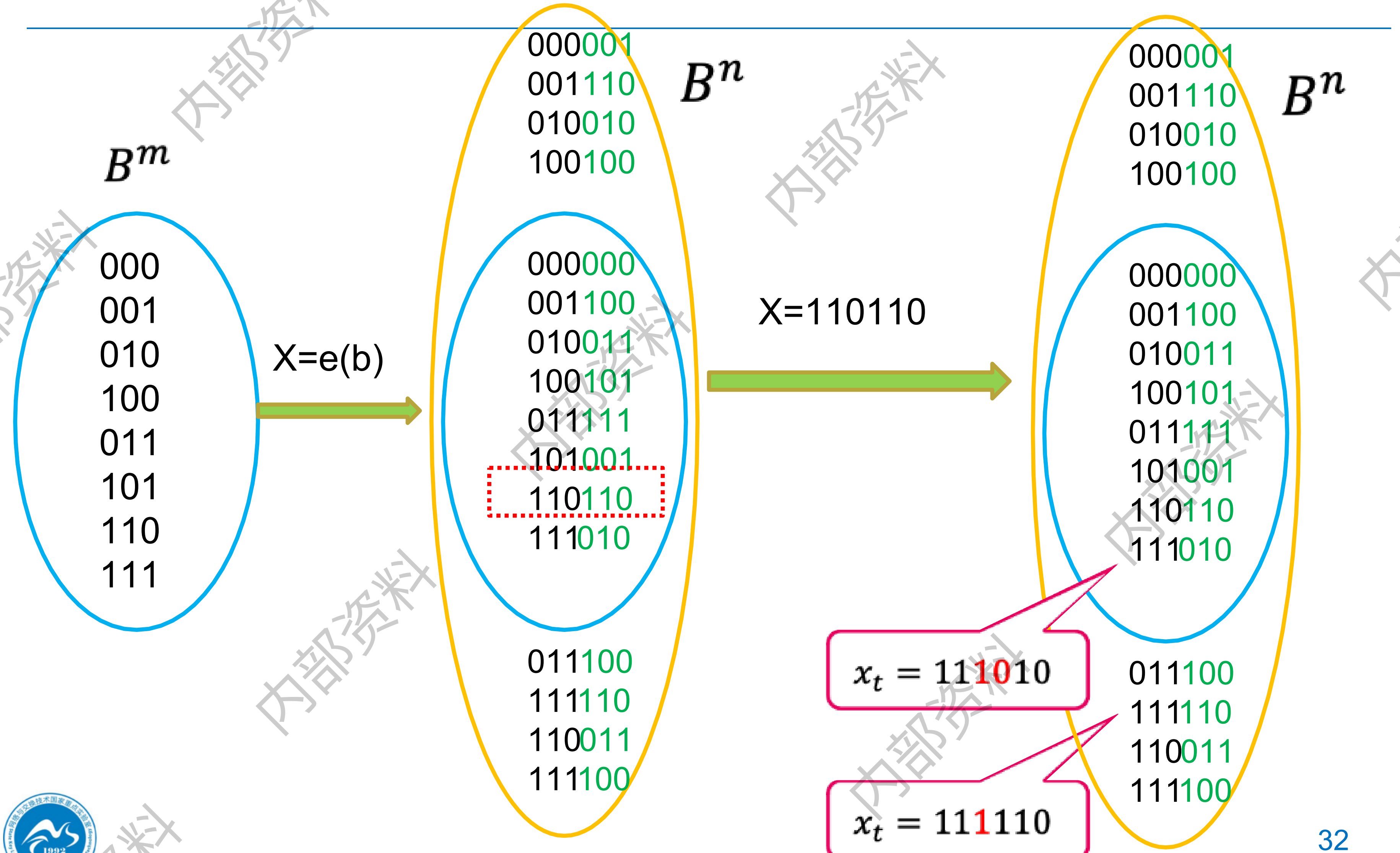
If $x = y$, that is, if x is transmitted and is mistakenly received as y, then $r \le k$ errors have been committed and have not been detected.

Thus it contradict with e can detect k or fewer errors.

Q.E.D.

思路--反证明：假设$e(b)$最短距离为r，且$r \le k$。

如果$\delta(x，x_t) \le k$，则无法检测k位错误。


B^m
000
001
010
100
011
101
110
111
X=e(b)
B^n
000001
001110
010010
100100
000000
001100
010011
100101
011111
101001
110110
111010
011100
111110
110011
111100
X=110110
B^n
000001
001110
010010
100100
000000
001100
010011
100101
011111
101001
110110
111010
011100
111110
110011
111100
$x_t = 111010$
$x_t = 111110$

Example 6

Consider the (3, 8) encoding function $e: B^3 \to B^8$ defined by

$$\delta_{\min} = 3 \qquad \left.\begin{aligned}
e(000) &= 00000000\\
e(001) &= 10011100\\
e(010) &= 00101101\\
e(011) &= 10010101\\
e(100) &= 10100100\\
e(101) &= 10001001\\
e(110) &= 00011100\\
e(111) &= 00110001
\end{aligned}\right\}\ \text{code word}$$

Group Codes – 群码

An (m, n) encoding function $e: B^m \to B^n$ is called a *group code* if

$$e(B^m) = \{e(b) \mid e(b) \in B^n\} = \mathrm{Ran}(e)$$

is a subgroup of B^n

Example 7: is e a group code?

Consider the (3, 6) encoding function e: $B^3 \rightarrow B^6$ defined by

验证是B^6的子群

$$
\left.
\begin{array}{l}
e(000) = 000000 \\
e(001) = 001100 \\
e(010) = 010011 \\
e(011) = 011111 \\
\hline
e(100) = 100101 \\
e(101) = 101001 \\
e(110) = 110110 \\
e(111) = 111010
\end{array}
\right\}\ \text{code word}
$$

Example 7: is e a group code?

We must show that the set of all code words

N = {000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010}

is a subgroup of B^6.

Theorem 3

Let e: $B^m \to B^n$ be a group code. The minimum distance of e is the minimum weight of a nonzero code word.

e 的最短距离是非零代码字的最小权。

$$\left.\begin{array}{l} e(000)=000000 \\ e(001)=001100 \\ e(010)=010011 \\ e(011)=011111 \\ \hline e(100)=100101 \\ e(101)=101001 \\ e(110)=110110 \\ e(111)=111010 \end{array}\right\} \text{code word}$$

Proof of Theorem 3

Let δ_{min} be the minimum distance of the group code, and suppose that $\delta_{min} = \delta(x, y)$, where x and y are distinct code words.

Also, let η be the minimum weight of a nonzero code word and suppose that $\eta = |z|$ for a code word z.

- 设e的最小距离$\delta_{min} = \delta(x, y)$
- 设z有最小权： $\eta = |z|$

Proof of Theorem 3

Since e is a group code, $x \oplus y$ is a nonzero code word. Thus

- $\delta_{min} = \delta(x, y) = |x \oplus y| \geq \eta$.——群封闭性$x \oplus y$为其中的一个代码字

On the other hand, since **0** and z are distinct code words,

- $\eta = |z| = |z \oplus \mathbf{0}| = \delta(z, \mathbf{0}) \geq \delta_{min}$

Hence $\eta = \delta_{min}$.

Q.E.D.

Example 8

The minimum distance of the group code in

$$\left.\begin{aligned} e(000) &= 000000 \\ e(001) &= 001100 \\ e(010) &= 010011 \\ e(011) &= 011111 \\ \hline e(100) &= 100101 \\ e(101) &= 101001 \\ e(110) &= 110110 \\ e(111) &= 111010 \end{aligned}\right\} \text{code word}$$

权值=2

δ_{min} =2

To check this directly would require 28 different calculations.

Constructing

group code

A review on Boolean matrices

- mod-2 sum D ⊕ E （模2布尔和）
- mod-2 Boolean product D ∗ E （模2布尔积）

EXAMPLE 9 We have

$$\begin{bmatrix}1&0&1&1\\0&1&1&0\\1&0&0&1\end{bmatrix} \oplus \begin{bmatrix}1&1&0&1\\1&1&0&1\\0&1&1&1\end{bmatrix} = \begin{bmatrix}1+1&0+1&1+0&1+1\\0+1&1+1&1+0&0+1\\1+0&0+1&0+1&1+1\end{bmatrix}$$

$$= \begin{bmatrix}0&1&1&0\\1&0&1&1\\1&1&1&0\end{bmatrix}$$

Let $\mathbf{D} = [\, d_{ij} \,]$ be an $m \times p$ Boolean matrix, and let $\mathbf{E} = [\, e_{ij} \,]$ be a $p \times n$ Boolean matrix. We define the **mod-2 Boolean product $\mathbf{D} * \mathbf{E}$** as the $m \times n$ matrix $\mathbf{F} = [\, f_{ij} \,]$, where

$$f_{ij} = d_{i1} \cdot e_{1j} + d_{i2} \cdot e_{2j} + \cdots + d_{ip} \cdot e_{pj}, \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

This type of multiplication is illustrated in Figure 11.3. Compare this with similar figures in Section 1.5.

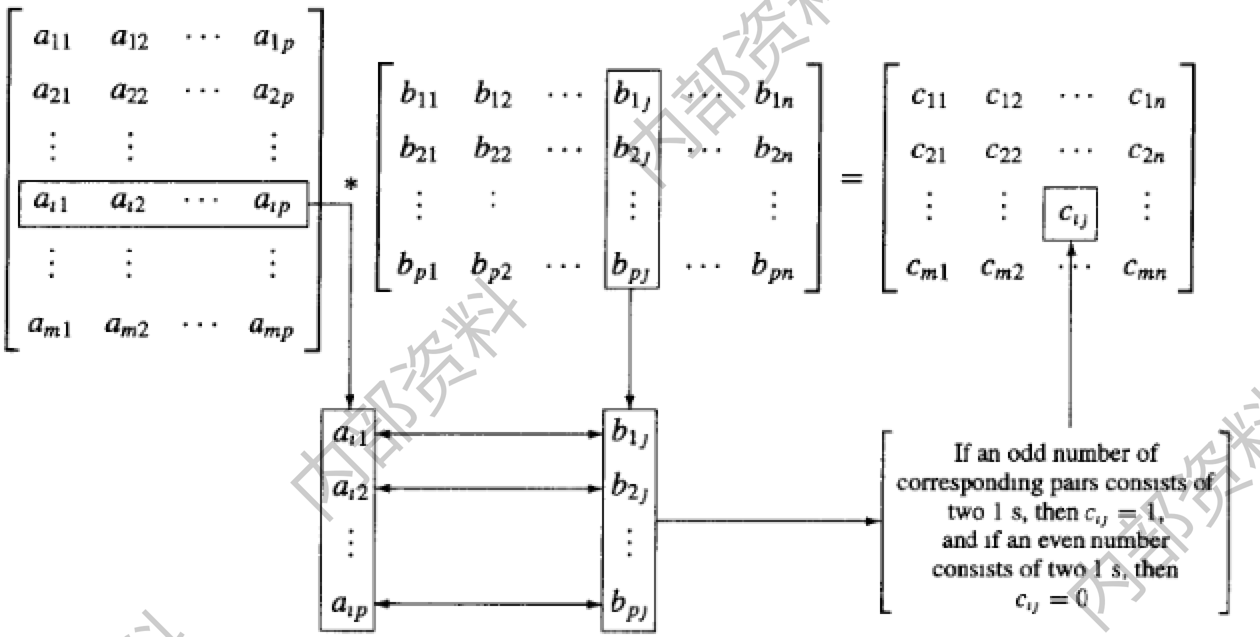


Figure 11.3

EXAMPLE 10 We have

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 1\cdot 1 + 0\cdot 0 & 1\cdot 0 + 1\cdot 1 + 0\cdot 1 \\ 0\cdot 1 + 1\cdot 1 + 1\cdot 0 & 0\cdot 0 + 1\cdot 1 + 1\cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Theorem 4

Let **D** and **E** be $m \times p$ Boolean matrices,
and let **F** be a $p \times n$ Boolean matrix. Then

$$(\mathbf{D} \oplus \mathbf{E}) * \mathbf{F} = (\mathbf{D} * \mathbf{F}) \oplus (\mathbf{E} * \mathbf{F}).$$

That is, a distributive property holds for $\oplus$ and $*$.

运算$*$ 对$\oplus$满足分配律

Notation

We shall now consider the element

$x = x_1x_2\ldots x_n \in B^n$

as the $1 \times n$ matrix

$[x_1\, x_2\, \ldots\, x_n\,]$

Theorem 5-同态

Let m and n be nonnegave integers with m<n, r=n-m,and Let H be an n×r Boolean matrix. Then the function f_H:$B^n \rightarrow B^r$ defined by

$f_H(x) = x * H, x \in B^n$

is a homomorphism from the group B^n to the group B^r

Proof

If x and y are elements in B^n, then

$$f_H(x \oplus y) = (x \oplus y) * \mathbf{H}$$
$$= (x * \mathbf{H}) \oplus (y * \mathbf{H}) \quad \text{by Theorem 4}$$
$$= f_H(x) \oplus f_H(y).$$

Hence f_H is a homomorphism from B^n to B^r

Q.E.D.

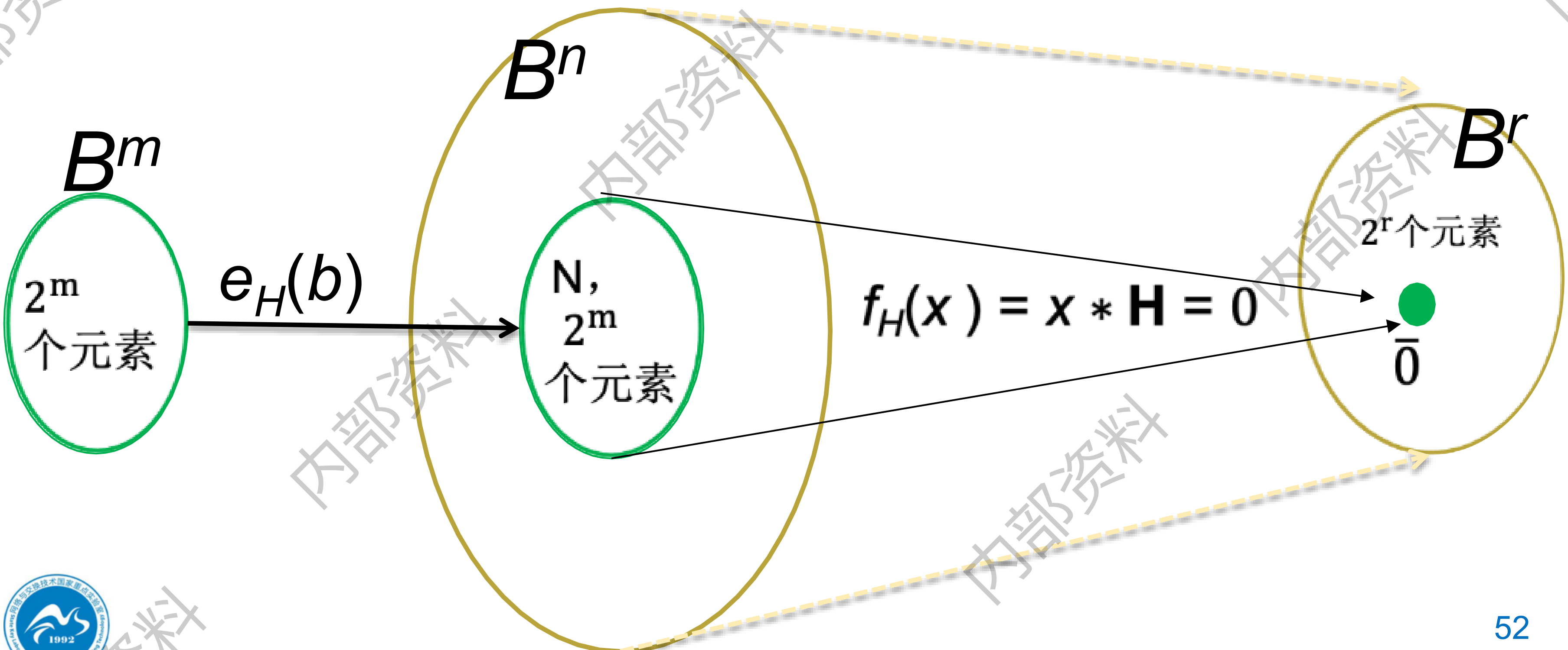
Corollary 1
推论1-正规子群

Let m, n, r, H, and f_H be as in Theorem 5.
Then
$N = \{x \in B^n \mid x * H = \overline{0}\}$
is a normal subgroup of B^n.

Corollary 1
推论1-证明

Proof:

N is the kernel of the homomorphism f_H, so it is a normal subgroup of B^n.



$$H=\begin{bmatrix}1&1&0\\0&1&1\\1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

B^5

N

00000	01011	10110	11101
01000	00011	11110	10101
10000	11011	00110	01101
11000	10011	01110	00101
00001	01010	10111	11100
00010	01001	10100	11111
10001	11010	00111	01100
10010	11001	00100	01111

B^3

000
011
110
101
001
010
111
100

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Parity check matrix – 一致性检验矩阵

Let $m < n$ and $r = n - m$, the following $n \times r$ Boolean matrix is called a *parity check matrix*.

$$H = \begin{bmatrix} H_{m\times r} \\ I_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Encoding function

Define an encoding function e_H: $B^m \rightarrow B^n$

If $b = b_1 b_2 \ldots b_m$,

let $x = e_H(b) = b_1 b_2 \ldots b_m \, x_1 x_2 \ldots x_r$, where

$$
\begin{aligned}
x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} \\
x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} \\
&\vdots \\
x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr}
\end{aligned} \qquad (1)
$$

$e_H: B^m \rightarrow B^n$ in matrix format

$b_{11}\ b_{12}\ b_{13} \ldots b_{1m} \Rightarrow b_{11}\ b_{12}\ b_{13} \ldots b_{1m}\ x_{11}\ x_{12} \ldots x_{1r}$

$$e_{\mathrm{H}}(B^m) = B^m * \begin{bmatrix} \mathrm{I}_m & \mathrm{H}_{m\times r} \end{bmatrix}$$

$$= \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & & \vdots \\ b_{2^m 1} & b_{2^m 2} & \cdots & b_{2^m m} \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 & h_{11} & h_{12} & \cdots & h_{1r} \\ 0 & 1 & \cdots & 0 & h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & h_{m1} & h_{m2} & \cdots & h_{mr} \end{bmatrix}$$

$$= \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} & x_{11} & x_{12} & \cdots & x_{1r} \\ b_{21} & b_{22} & \cdots & b_{2m} & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ b_{2^m 1} & b_{2^m 2} & \cdots & b_{2^m m} & x_{2^m 1} & x_{2^m 2} & \cdots & x_{2^m r} \end{bmatrix}$$

例：4个校验位

1	0	1	1	0	0	0	1	1	0	1	0	0	0

1	0	1	1	0	1	0	1	1	0	1	0	0	0

b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 x11 x12 x13 x14

b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 x11 x12 x13 x14

b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 x11 x12 x13 x14

b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 x11 x12 x13 x14

$$\mathrm{x11} = b1 \oplus b3 \oplus b5 \oplus b7 \oplus b9$$
$$\mathrm{x12} = b2 \oplus b3 \oplus b6 \oplus b7 \oplus b10$$
$$\mathrm{x13} = b4 \oplus b6 \oplus b7$$
$$\mathrm{x14} = b8 \oplus b9 \oplus b10$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 \cdots & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 \ddots & 0 & 1 \\ & \cdots & & \end{bmatrix}$$

Theorem 6

Let x=$y_1y_2 \ldots y_\mathrm{m}x_1 \ldots x_r \in B^\mathrm{n}$ .Then

x∗H=$\overline{0}$

if and only if

x=e_H(b) for some $b \in B^m$

Proof: $x * \mathbf{H} = \bar{0} \Rightarrow x = e_H(b)$ for some $b \in B^m$

$x = y_1 y_2 \ldots y_m x_1 \ldots x_r$

Suppose that $\boldsymbol{x} * \boldsymbol{H} = \bar{0}$

$$y_1 \cdot h_{11} + y_2 \cdot h_{21} + \ldots + y_m \cdot h_{m1} + x_1 = 0$$
$$y_1 \cdot h_{12} + y_2 \cdot h_{22} + \ldots + y_m \cdot h_{m2} + x_2 = 0$$
$$\vdots$$
$$y_1 \cdot h_{1r} + y_2 \cdot h_{2r} + \ldots + y_m \cdot h_{mr} + x_r = 0$$

$$\mathrm{H} = \begin{bmatrix} \mathrm{H}_{m\times r} \\ \mathrm{I}_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Proof: $x * \mathbf{H} = \bar{0} \Rightarrow x = e_H(b)$ for some $b \in B^m$

Note that x_i +x_i=$\bar{0}$. So add x_i to i^{th} row and get

$$x_i = y_1 \cdot h_{1i} + y_2 \cdot h_{2i} + \ldots + y_m \cdot h_{mi}$$

Letting $b_1 = y_1$, $b_2 = y_2$,⋯, $b_m = y_m$, We see that x_1,x_2,… , x_r, satisfy the equations in (I).

Thus b= $b_1, b_2, \ldots, b_m \in B^m$ and x=e_H(b)

$$\begin{aligned} x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} \\ x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} \\ &\vdots \\ x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr} \end{aligned}$$

Proof: $x * \mathbf{H} = \bar{0} \Leftarrow x = e_H(b)$ for some $b \in B^m$

对右式两边模加x_i： $x_i = y_1 \cdot h_{1i} + y_2 \cdot h_{2i} + \ldots + y_m \cdot h_{mi}$

Conversely if x=e_H(b) , the equations In (I) can be rewritteri by adding x_i to both sides of the i^{th} equation, *i*=1, 2, …,*n*, as

$$\begin{aligned}
&b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} + x_1 = 0 \\
&b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} + x_2 = 0 \\
&\vdots \\
&b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr} + x_r = 0
\end{aligned}$$

which shows $\boldsymbol{x} * \boldsymbol{H} = \bar{\mathbf{0}}$

Q. E. D.

Corollary 2

$e_H(B^m) = \{e_H(b) \mid b \in B^m\}$ is a subgroup of B^n

Proof :

The result follows from the observation that

$e_H(B^m) = \ker(f_H)$

and from Corollary I.

Thus e_H is a group code.

Example 11

Let $m = 2$, $n = 5$, and

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Determine the group code e_H: $B^2 \to B^5$.

Solution 11

$$e(B^m) = B^m * \mathrm{H}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

编码结果

e(00)=00000

e(01)=01011

e(10)=10110

e(11)=11101

Homework

22,26,30 @ 11.1

练习

Let

$$H=\begin{bmatrix}0 & 1 & 1\\ 0 & 1 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

be a parity check matrix. Determine the (2, 5) group code function e_H: $B^2 \to B^5$.

e(B^2)=$B^2 * H$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Decoding and Error Correction

译码与纠错

Decoding Function – 译码函数

Consider an (m, n) encoding function

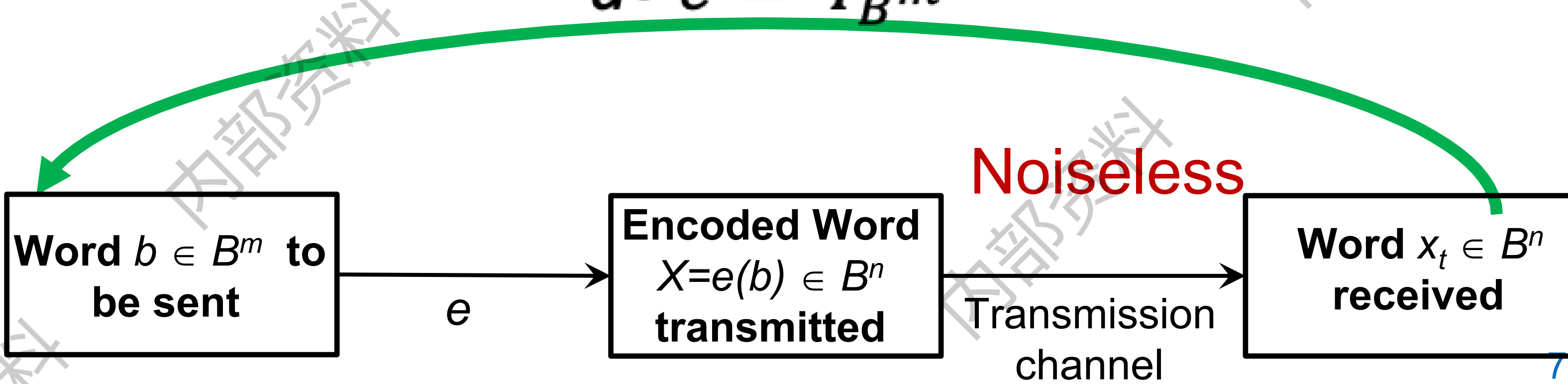
$e: B^m \to B^n$.

Once the encoded word $x = e(b) \in B^n$, for $b \in B^m$, is received as the word x_t, we are faced with the problem of identifying the word b that was the original message.

Decoding Function

Consider an (m, n) encoding function
$e: B^m \to B^n$.
An onto function $d: B^n \to B^m$ is called an (n,m) decoding function associated with e(与e关联的译码函数） if $d(x_t) = b' \in B^m$ is such that when the transmission channel has no noise, then b' = b, that is,

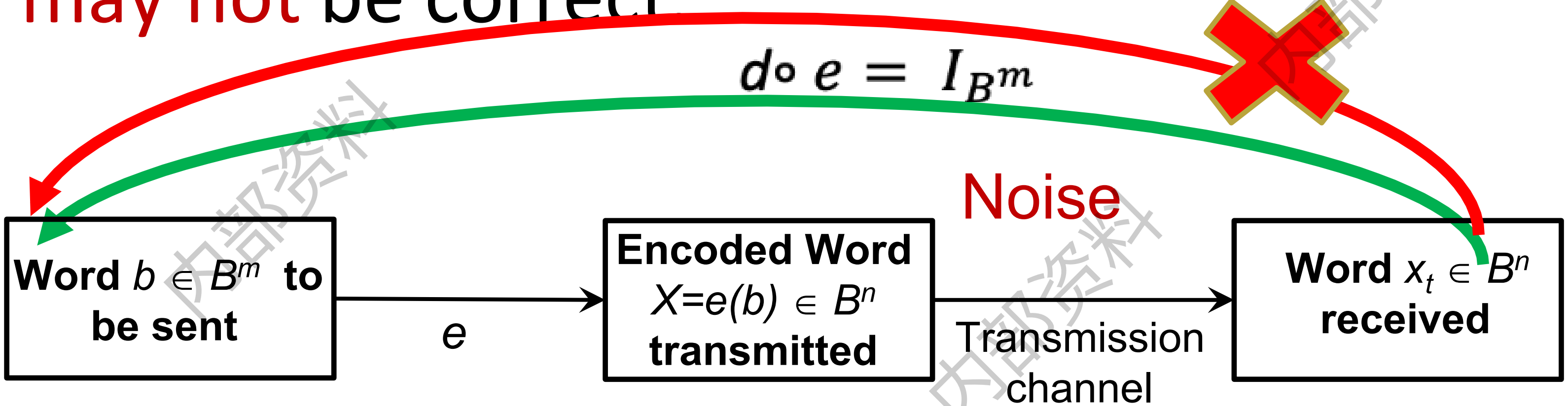
$$d \circ e = I_{B^m}$$



Decoding Function

The decoding function *d* is required to be onto so that every received word can be decoded to give a word in B^m.

It decodes properly received words correctly, but the decoding of improperly received words may or may not be correct.



Example 1

§11.1- exmaple 1， e： $B^m \to B^{m+1}$

Define the decoding function d: $B^{m+1} \to B^m$.

If $y = y_l y_2 \ldots y_m y_{m+l} \in B^{m+1}$, then $d(y) = y_l y_2 \ldots y_m$

Observe that if $b = b_l b_2 \ldots b_m \in B^m$, then

$(d \circ e)(b) = d(e(b)) = b$

so $d \circ e = \mathrm{I}_{Bm}$

For a concrete example, let $m = 4$.

Then $d(10010) = 1001$ and $d(11001) = 1100$

Decoding function

Let e be an (m, n) encoding function and let d be an (n, m) decoding function associated with e.

The pair (e, d) is said to *correct k or fewer errors*

if whenever $x = e(b)$ is transmitted correctly or with k or fewer errors and x_t is received, then $d(x_t) = b$. Thus x_t is decoded as the correct message b.

Example 2

Consider the $(m, 3m)$ encoding function defined in Example 3 of Section 11.1. Define the decoding function $d: B^{3m} \to B^m$.

Let $y = y_1y_2\ldots y_my_{m+1}\cdots y_{2m}y_{2m+1}\cdots y_{3m}$, Then

$d(y) = z_1z_2\ldots z_m$

Where

$$z_i = \begin{cases} 1 & \text{if}\{y_i, y_{i+m}, y_{i+2m}\} \text{ has at least two } 1'\text{s} \\ 0 & \text{if}\{y_i, y_{i+m}, y_{i+2m}\} \text{ has less than two } 1'\text{s} \end{cases}$$

E.g. x_t = 011 011 111, then $d(x_t)$ = 011

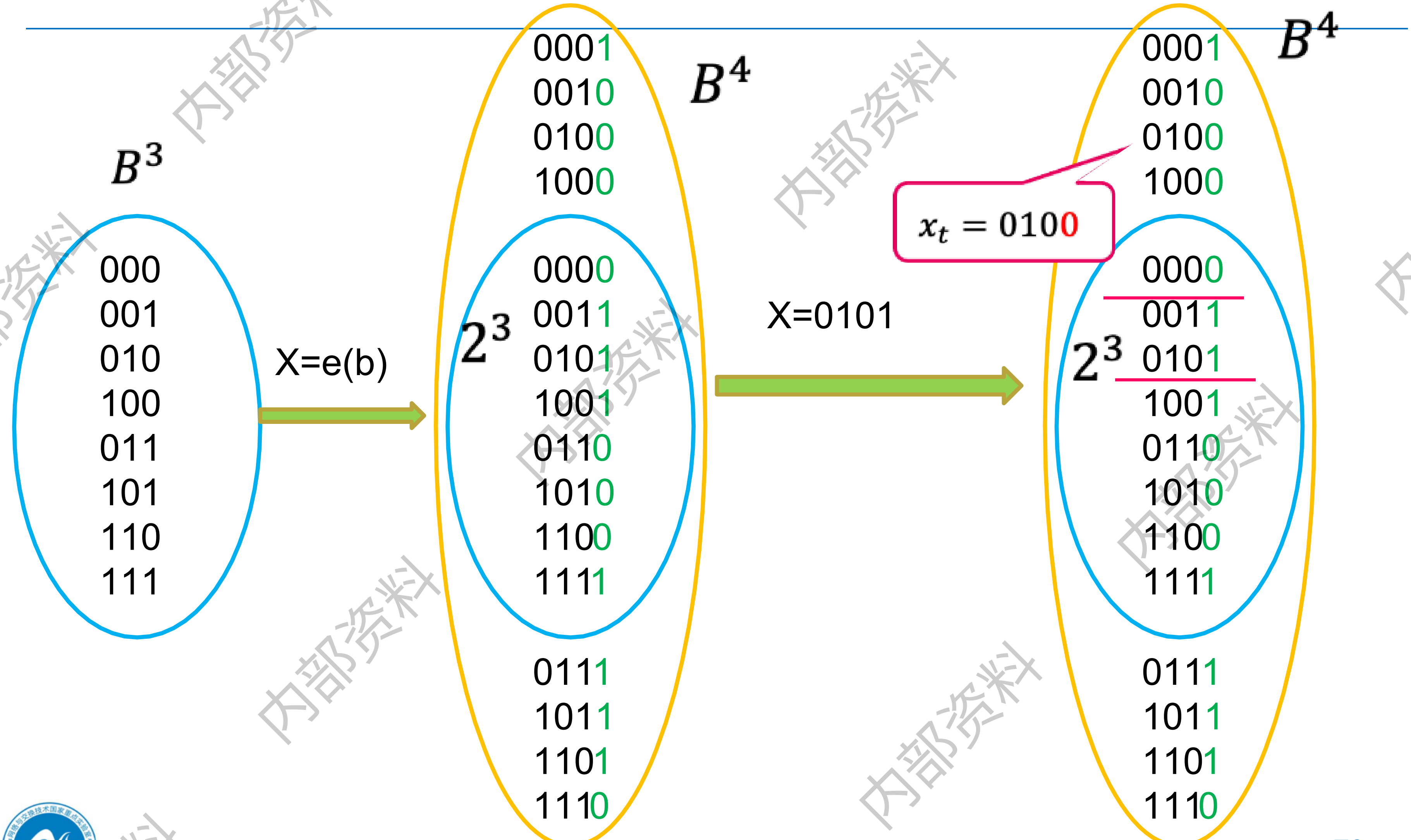
Maximum likelihood technique – 最大似然技术

Since B^m has 2^m elements, there are 2^m code words in B^n. List it as

$x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}$

If the received word is x_t, we compute $\delta(x^{(i)}, x_t)$ for $1 \leq i \leq 2^m$, and choose the first code word, say it is $x^{(s)}$, such that

$$\min_{1 \leq i \leq 2^m} \{\delta(x^{(i)}, x_t)\} = \delta(x^{(s)}, x_t)$$


B^3
000
001
010
100
011
101
110
111
X=e(b)
B^4
0001
0010
0100
1000
2^3
0000
0011
0101
1001
0110
1010
1100
1111
0111
1011
1101
1110
X=0101
B^4
$x_t = 0100$
0001
0010
0100
1000
2^3
0000
0011
0101
1001
0110
1010
1100
1111
0111
1011
1101
1110

Maximum likelihood decoding function

That is, $x^{(s)}$ is a code word that is closest to x_t and the first in the list.

If $x^{(s)} = e(b)$, we define the *maximum likelihood decoding function* d associated with e by

$d(x_t) = b$

$$\delta_{min} = \delta\left(x_t, x^{(s)}\right)$$
$$\Longrightarrow \quad d(x_t) = b$$

x =b

$x_t = b^i$

$b_1^{(1)}\ b_2^{(1)}\ b_3^{(1)} \ldots\ b_m^{(1)} \ldots\ b_n^{(1)}$

$b_1^{(2)}\ b_2^{(2)}\ b_3^{(2)} \ldots\ b_m^{(2)} \ldots\ b_n^{(2)}$

…………

$b_1^{(i)}\ b_2^{(i)}\ b_3^{(i)} \ldots\ b_m^{(i)} \ldots\ b_n^{(i)}$

$\delta(x^{(s)}, x_t)$

Min δ

…………

$b_1^{(j)}\ b_2^{(j)}\ b_3^{(1)} \ldots\ b_m^{(j)} \ldots\ b_n^{(j)}$

…………

$b_1^{(2^m)}\ b_2^{(2^m)}\ b_3^{(2^m)} \ldots\ b_m^{(2^m)} \ldots b_n^{(2^m)}$

Theorem 1

Suppose that e is an (m, n) encoding function and d is a maximum likelihood decoding function associated with e. Then

(e, d) can correct k or fewer errors

if and only if

the minimum distance of e is at least $2k + 1$.

$$\forall x, y, (e, d)\ correct\ errors \le \mathrm{k},$$
$$\text{iff},\ \ \delta(x, \mathrm{y}) \ge 2k + 1$$

Proof(1/3)

- *由e的最短距离至少2k+1，证能纠错至多k位*

（1）2k+1 ≤ $\delta(x,z) \le \delta(x, x_t)+\delta(x_t,z) \le$ k+$\delta(x_t,z)$

所以，$\delta(x_t,z) \ge$2k+1-k=k+1.说明x_t与x更接近，

故d（x_t）=b

（2）假设代码字之间的最短距离 r ≤ 2k

设 x=e(b) 和 x'=e(b') 是编码后的两个代码字

Proof(2/3)

(a)当r ≤ k时，如下图：

$$x_t = b'_1\ b'_2 \ldots b'_r\ b'_{r+1} \ldots b'_k \ldots b_n$$

发送端　　　　　　　　　　　　接收端x_t

$$x = b_1\ b_2 \ldots\ b_r\ b_{r+1} \ldots\ b_k \ldots b_n$$

$$x_t = b'_1\ b'_2 \ldots b'_r\ b'_{r+1} \ldots b'_k \ldots b_n$$

$$x' = b'_1\ b'_2 \ldots b'_r\ b_{r+1} \ldots b_k \ldots b_n$$

…………

$\delta(x\ \ ,x_t)$=k

$\delta(x',x_t)$=k-r

Proof（3/3)

（2）假设代码字之间的最短距离 $k+1 \le r \le 2k$

(b)当$k+1 \le r \le 2k$时，

代码字x_t与x' 的距离 $\delta(x_t, x')=r-k \le k$

$$x = b_1 \ b_2 \ldots \ldots b_k \ b_{k+1} \ldots\ldots b_r \ b_{r+1} \ldots b_{2k} \ldots\ldots b_n$$

$$x_t = b'_1 \ b'_2 \ldots \ b'_k \ b_{k+1} \ldots b_r \ b_{r+1} \ldots b_{2k} \ldots b_n$$

$$x' = b'_1 \ b'_2 \ldots b'_k \ b'_{k+1} \ldots b'_r \ b_{r+1} \ldots b_{2k} \ldots\ldots b_n$$

Example 3

How many errors can (e, d) correct?

The (3, 8) encoding function $e: B^3 \to B^8$

$$\left.\begin{aligned}
e(000) &= 00000000\\
e(001) &= 10011100\\
e(010) &= 00101101\\
e(011) &= 10010101\\
\hline
e(100) &= 10100100\\
e(101) &= 10001001\\
e(110) &= 00011100\\
e(111) &= 00110001
\end{aligned}\right\} \text{code word}$$

$\delta_{min} = 3,\quad 3 \geq 2k+1 \;\Rightarrow k \leq 1$

and let d be an (8, 3) maximum likelihood decoding function associated with e.

Constructing

maximum likelihood decoding function associated with a given group code

Theorem 2

If *K* is a finite subgroup of a group *G*, then every left coset of *K* in *G* has exactly as many elements as *K*.

subgroup :$K \subseteq G$, $|K| = |aK|$

Recall

A finite group can be represented in the form of the multiplication table.

$ax = b$

$ya = b$

*	a1	a2	…	an
a1				
a2				
…			$a_i * a_j$	
an				

Proof

Let aK be a left coset of K in G, where $a \in G$. Consider the function f: $K \to aK$ defined by

- $f(k) = ak$, for $k \in K$. We show that f is one to one and onto.

To show that f is one to one, we assume that

- $f(k_l) = f(k_2)$, $k_l, k_2 \in K$. （证单射）

Then $ak_l = ak_2$

Left multiply a^{-1}, we have

- $a^{-1}\, ak_l = a^{-1}\, ak_2$
- $k_l = k_2$

Proof

To show that f is onto, let b be an arbitrary element in aK.

Then $b = ak$ for some $k \in K$. We now have

$$f(k) = ak = b$$

so f is onto.

Since f is one to one and onto, K and aK have the same number of elements.

Q.E.D

Group code word
群码字

Let e: $B^m \rightarrow B^n$ be an (m, n) encoding function that is a group code.

Thus the set N of code words in B^n is a subgroup of B^n whose order is 2^m, say

$N = \{x^{(1)}, x^{(2)},\dots, x^{(2^m)}\}$.

$$N \subset B^n$$

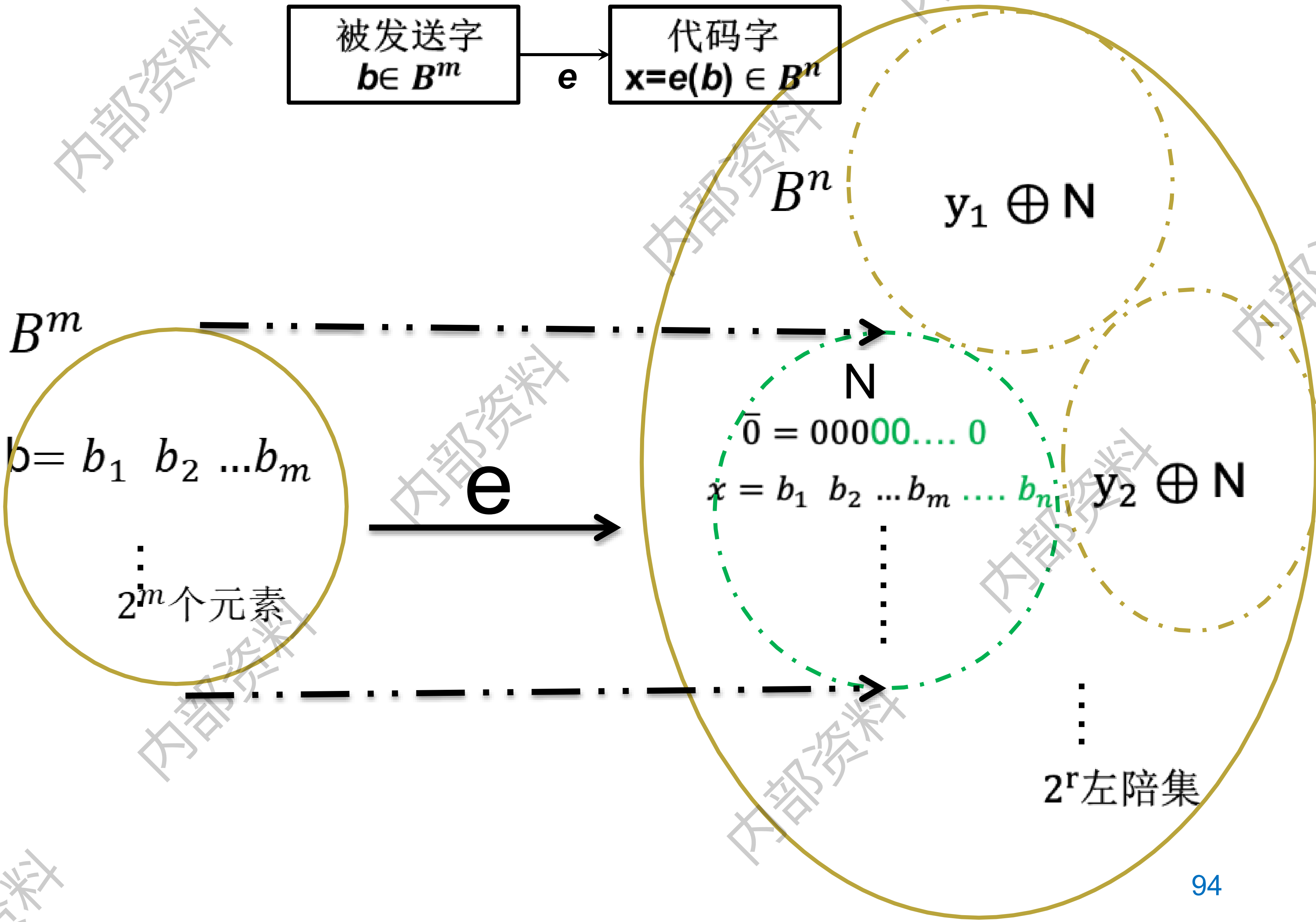
N是群B^n 子群，假设N中包括如下的元素

$N=\{\bar{0}\ ,\ x^{(2)}, x^{(3)} \ldots , x^{(i)} .. , x^{(j)} \ldots, x^{(2^m)}\}$

在N之外任意取一个元素y，则y ⊕N形成一个左陪集

$y \oplus N=\{y \oplus \bar{0}\ , y \oplus x^{(2)}, y \oplus x^{(3)} \ldots , y \oplus x^{(i)} .. , y \oplus x^{(j)} \ldots, y \oplus x^{(2^m)}\}$

除N之外，共可生成2^r-1 个左陪集


被发送字
b∈ B^m
e
代码字
x=e(b) ∈ B^n
B^n
$y_1 \oplus$ N
B^m
N
$\bar{0} = 00000....0$
b= $b_1\ b_2\ ...b_m$
e
$x = b_1\ b_2\ ...b_m\ ....\ b_n$
$y_2 \oplus$ N
2^m个元素
2^r左陪集

回顾陪集

例$(Z_6,\oplus)$,子群H={[0],[2],[4]}

[0] H={[0]⊕[0],[0]⊕[2],[0]⊕[4]}={[0],[2],[4]}

[1] H={[1]⊕[0],[1]⊕[2],[1]⊕[4]}={[1], [3], [5]}

[2] H={[2]⊕[0],[2]⊕[2],[2]⊕[4]}={[0],[2],[4]}

[3] H={[3]⊕[0],[3]⊕[2],[3]⊕[4]}={ [3], [5]， [1]}

[4] H={[4]⊕[0],[4]⊕[2],[4]⊕[4]}={[0],[2],[4]}

[5] H={[5]⊕[0],[5]⊕[2],[5]⊕[4]}={[5]， [1], [3]}

群Z_6等于所有陪集的并

左陪集个数$= |Z_6|/|H|$

Coset leader – 陪集头

$N=\{\bar{0}, x^{(2)}, x^{(3)}\ldots, x^{(i)}\ldots, x^{(j)}\ldots, x^{(2^m)}\}$

Suppose that the code word $x = e(b)$ is transmitted and that the word x_t is received.

The left coset of x_t is

- $x_t \oplus N = \{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{2m}\}$ where $\varepsilon_i = x_t \oplus x^{(i)}$

if ε_i is a coset member with smallest weight, then $x^{(i)}$ must be a code word that is closest to x_t.

An element ε_i , having smallest weight, is called a *coset leader*.

陪集元素重排

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	…	$x^{(i)}$	…	$x^{(j)}$	…	$x^{(2^m)}$
$x_t \oplus \bar{0}$ $=\varepsilon_0$	$x_t \oplus x^{(2)}$ $=\varepsilon_2$	$x_t \oplus x^{(3)}$ $=\varepsilon_3$		$x_t \oplus x^{(i)}$ $=\varepsilon_i$		$x_t \oplus x^{(j)}$ $=\varepsilon_j$		$x_t \oplus x^{(2^m)}$ $=\varepsilon_{2^m}$

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	…	$x^{(i)}$	…	$x^{(j)}$	…	$x^{(2^m)}$
$\varepsilon_i \oplus \bar{0}$	$\varepsilon_i \oplus x^{(2)}$	$\varepsilon_i \oplus x^{(3)}$		$\varepsilon_i \oplus x^{(i)}$		$\varepsilon_i \oplus x^{(j)}$		$\varepsilon_i \oplus x^{(2^m)}$

$x_t \oplus N = \{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{2m}\}$，其中$\varepsilon_i = x_t \oplus x^{(i)}$

$\delta_{\min} = |\varepsilon_i|$，说明$x_t$与$x^{(i)}$距离最近

$x_t \oplus \varepsilon_i = x_t \oplus x_t \oplus x^{(i)} = x^{(i)}$

即： x_t与陪集头运算，可以译码出$x^{(i)}$

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$$H=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

B^5

N

00000	01011	10110	11101
01000	00011	11110	10101
10000	11011	00110	01101
11000	10011	01110	00101
00001	01010	10111	11100
00010	01001	10100	11111
10001	11010	00111	01100
10010	11001	00100	01111

B^3

000
011
110
101
001
010
111
100

Procedure

For obtaining a maximum likelihood decoding function *d* associated with a given group code *e*: $B^m \to B^n$

Step 1: Determine all the left cosets of $N=e(B^m)$ in B^n

Step 2: For each coset, find a coset leader (a word of least weight) .

Step 3: If the word x_t, is received, determine the coset of N to which x_t belongs.

Step 4: Let ε be a coset leader for the coset determined in Step 3. Compute $x = x_t \oplus \varepsilon$. If $x=e(b)$, we let $d(x_t) = b$. That is, we decode x_t as b.

Decoding table

If we receive the word x_t, we locate it in the table. If x is the element of N that is at the top of the column containing x_t, then x is the code word closest to x_t.

if $x = e(b)$, then $d(x_t) = b$.

$\bar{0}$	$x^{(2)}$	$x^{(3)}$	…	$x^{(2^m-1)}$
$\varepsilon^{(2)}$	$\varepsilon^{(2)} \oplus x^{(2)}$	$\varepsilon^{(2)} \oplus x^{(3)}$	…	$\varepsilon^{(2)} \oplus x^{(2^m-1)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
$\varepsilon^{(2')}$	$\varepsilon^{(2')} \oplus x^{(2)}$	$\varepsilon^{(2')} \oplus x^{(3)}$	…	$\varepsilon^{(2')} \oplus x^{(2^m-1)}$

Example 4

Consider the (3, 6) group code

$N = \{000000,$
$001100,$
$010011,$
$011111,$
$100101,$
$101001,$
$110110,$
$111010\}$
$= \{x^{(1)}, x^{(2)}, \ldots, x^{(8)}\}.$

Example 4
Constructing decoding table:

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	111110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000

Example 4

Constructing decoding table:

X=100101 $x_t \neq 000101$ $x_t = 100001$

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	111110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
000110	001010	010101	011001	100011	101111	110000	111100
010100	011000	000111	001011	110001	111101	100010	101110

换陪集头重排序

Simplified decoding technique 简化译码技术

Suppose that the (m, n) group code is e_H: $B^m \to B^n$, where **H** is a given parity check matrix.

$$
H = \begin{bmatrix} H_{m\times r} \\ I_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}
$$

Simplified decoding technique
简化译码技术

Then the function f_H: $B^n \to B^r$ defined by

$$f_H(x) = x * \mathbf{H},\quad x \in B^n$$

is a homomorphism from the group B^n to the group B^r.

11.1 theorm 5

Theorem 3

If *m*, *n*, *r*, **H**, and f_H are as defined, then

f_H is onto.

-- f_H 是满射的

Proof

Let $b = b_1 b_2 \ldots b_r$ be any element in B^r.

（ ∀ b ∈ B^r）

B^n : Letting $x = 0_1 \ldots 0_m b_1 b_2 \ldots b_r$ （ ∃ x ∈ B^n ）

Then $x * \mathbf{H} = b$.

Thus $f_H(x) = b$, so f_H is onto.

Example 4 (3, 6)

$$H = \begin{bmatrix} H_{m\times r} \\ I_r \end{bmatrix} = \left[\begin{array}{cccc} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{array}\right]$$

Br ={
$\boxed{000}$
$\boxed{001}$
$\boxed{010}$
100
011
101
110
111 }

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	111110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
000110	001010	010101	011001	100011	101111	110000	111100
010100	011000	000111	001011	110001	111101	100010	101110

Syndrome –校验子

It follows from Corollary I of Section 9.5 that B^r and B^n/N are isomorphic, where

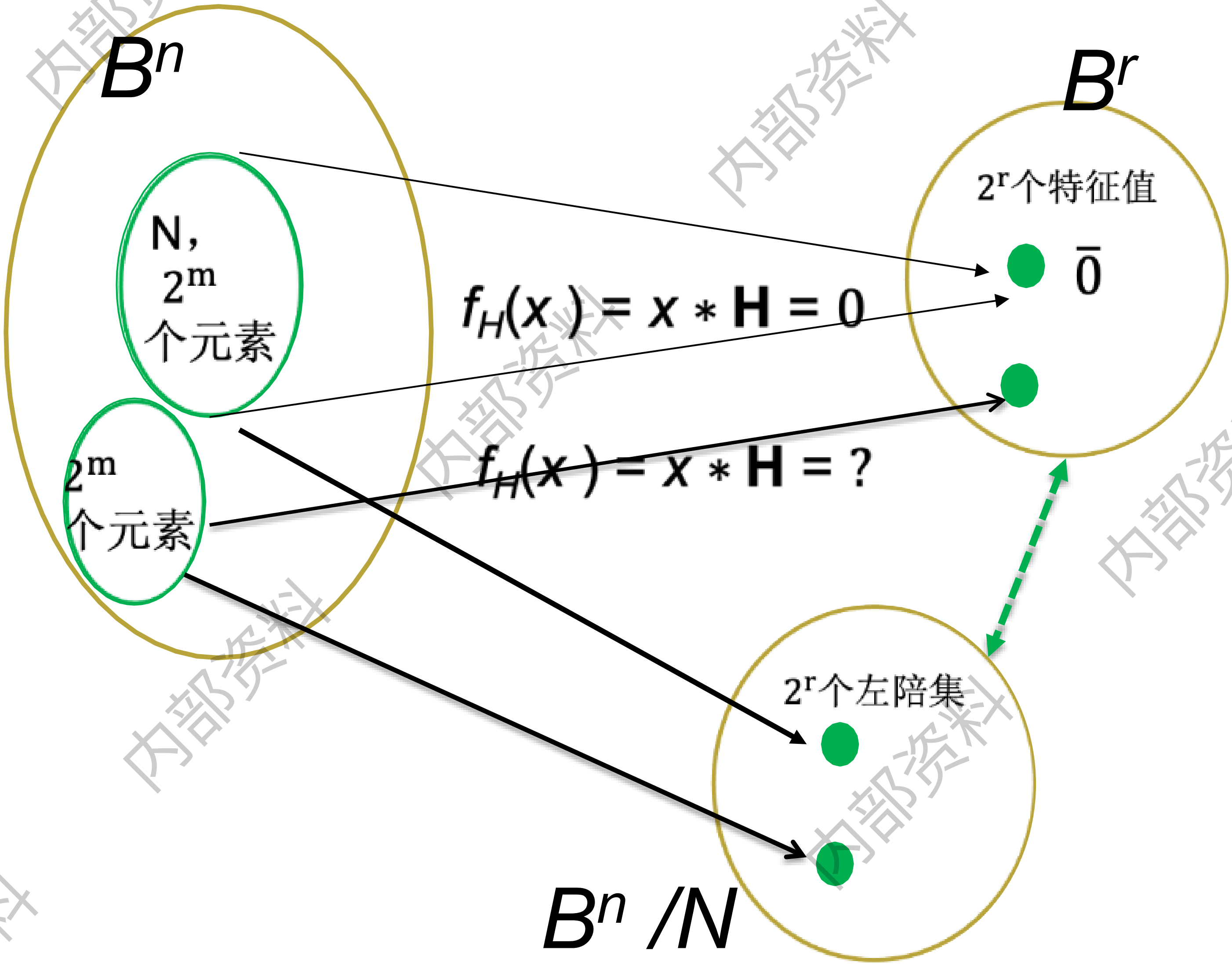
- $N = \{x \in B^n \mid x * \mathbf{H} = \mathbf{0}\} = \ker(f_H) = e_H(B^m)$

under the isomorphism

- g: $B^n/N \to B^r$ defined by
- $g(xN) = f_H(x) = x * \mathbf{H}$

The element x*H is called the syndrome （特征值或校验

9.5 Corollary 1，Br and Bn/N Isomorphism


B^n
N，
2^m
个元素
2^m
个元素
$f_H(x) = x * \mathbf{H} = 0$
$f_H(x) = x * \mathbf{H} = ?$
B^r
2^r个特征值
$\bar{0}$
2^r个左陪集
B^n/N

Theorem 4

Let x and y be elements in B^n. Then

x and y lie in the same left coset of N in B^n

if and only if

$f_H(x) = f_H(y)$

That is， if and only if

they have the same syndrome.

Proof

It follows from Theorem 4 of Section 9.5 that x and y lie in the same left coset of N in B^n if and only if $(-x) \oplus y \in N$. （ $a^{-1}b \in N \Longrightarrow$ a，b在一个陪集中）

$\Longrightarrow x \oplus y \in N$

Since $N = \ker(f_H)$,

$x \oplus y \in N$

if and only if

$f_H(x \oplus y) = 0_{B^r}$

$f_H(x) \oplus f_H(y) = 0_{B^r}$

$f_H(x) = f_H(y)$

Q.E.D.

$$H=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$B^2 \to B^5$

N

00000	01011	10110	11101
01000	00011	11110	10101
10000	11011	00110	01101
11000	10011	01110	00101
00001	01010	10111	11100
00010	01001	10100	11111
10001	11010	00111	01100
10010	11001	00100	01111

B^3

000
011
110
101
001
010
111
100

Decoding procedure

Suppose that we compute the syndrome of each coset leader. If the word x_t is received, we also compute $f_H(x_t)$, the syndrome of x_t. By comparing $f_H(x_t)$ and the syndromes of the coset leaders, we find the coset in which x_t lies. Suppose that a coset leader of this coset is ε. We now compute $x = x_t \oplus \varepsilon$. If $x = e(b)$, we then decode x_t as b.

New procedure

Step 1: Determine all left cosets of $N = e_H(B^m)$ in B^n.

Step 2: For each coset, find a coset leader, and compute the syndrome of each leader

Step 3: If x_t is received, compute the syndrome of x_t and find the coset leader ε having the same syndrome. Then $x_t \oplus \varepsilon = x$ is a code word $e_H(b)$, and $d(x_t) = b$.

Example 5

Consider the parity check matrix and the (3, 6) group code e_H: $B^3 \to B^6$.

$$
H = \begin{bmatrix} 110 \\ 101 \\ 011 \\ 100 \\ 010 \\ 001 \end{bmatrix}
\qquad
\left.\begin{array}{l}
e(000) = 000000 \\
e(001) = 001011 \\
e(010) = 010101 \\
e(011) = 011110 \\ \hline
e(100) = 100110 \\
e(101) = 101101 \\
e(110) = 110011 \\
e(111) = 111000
\end{array}\right\} \text{code word}
$$

Example 5

N = {000000, 001011, 010101, 011110, 100110, 101101, 110011, 111000}

```
Syndrome of Coset Leader    |       Coset leader
----------------------------+-------------------
           000              |      000000
           001              |      000001
           010              |      000010
           011              |      001000
           100              |      000100
           101              |      010000
           110              |      100000
           111              |      001100
----------------------------+-------------------
```

Example 5

If x_t = 001110, then $f_H(x_t) = x_t * \mathbf{H}$ = 101, same as ε=010000.

$x = x_t \oplus \varepsilon$ = 001110 $\oplus$ 01000 = 011110 = e(011), so decode 001110 as 011.

Syndrome of Coset Leader	Coset leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

$x_t * \mathbf{H}$ = syndrome

Homework

16, 20, 26， 28 @11.2

Key ideas for review

Message, word

(m, n) encoding function, one-to-one

- *Code word, parity check code*
- *Detect, correct, k or fewer errors*

Hamming distance

- *Properties of distance*

Group code and parity check matrix

- *Minimum distance of a group code*

Maximum likelihood technique

- *Maximum likelihood decoding function*
- *Syndrome and decoding procedure for group code*

Thank you for your attention!